

Measurement Science

Science

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Short Title:	Measurement Science	
Faculty	Science and Computing	
Department	Science	
Cluster	Mathematics and Physics	
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Credits	10	Level: Introductory

Description of Module / Aims

This is an introductory module in measurement science. Much of the content is basic physics, providing the student with a background to understand the more applied measurement concepts considered towards the end of the module. No previous knowledge of physics or measurement science is assumed.

Programmes

PHYS-0018 Higher Certificate in Science in Good Manufacturing Practice and Technology 1 / 2 / M
(WD_SGMPT_C)

Indicative Content

- Introduction: SI system of units; scientific notation; simple calculations using physical quantities
- Basics mechanics: vectors and scalars; Newtons laws of motion and gravitation
- Heat and temperature: specific heat capacity and latent heat
- Waves motion and light: electromagnetic waves and electromagnetic spectrum
- Introduction to electricity: simple circuits; resistors in series and parallel; electrical safety
- Analogue and digital signals: temperature sensors
- Measurement science: systematic and random errors; strain gauge and other transducers; calibration, standards, traceability ladder

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use the SI system of units and perform standard types of formula-based calculations in basic physics.
2. Define basic terms in physics and describe applications of the basic physics presented.
3. Explain the principles behind a wide range of temperature sensors.
4. Calculate important quantities associated with simple electrical circuits.
5. Use laboratory equipment in a safe and accurate manner.
6. Describe the basic components of a measurement system and outline important measurement principles.
7. Report on and critically assess experiments carried out to demonstrate principles of physics in a laboratory context.

Learning and Teaching Methods

- Lectures.
- Practical programme.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,6
Continuous Assessment	50%	
<i>Practical</i>	50%	5,7

Assessment Criteria

- <40%:** The student has failed to demonstrate a reasonable grasp of the material and is unable to carry out routine calculations reliably. Experimental work is generally poor.
- 40%-49%:** Able to demonstrate a basic understanding of the fundamental concepts outlined in the indicative content. Able to provide partial solutions to numerical problems. Practical work is usually of an acceptable quality.
- 50%-59%:** Able to clearly describe and define concepts encountered in the indicative content. Able to analyse physical situations and apply suitable problem-solving techniques to find numerical solutions. Carries out practical work capably and writes up same to an acceptable standard.
- 60%-69%:** In addition to the above, able to apply problem-solving techniques to new similar problems. Demonstrates a good understanding of the majority of the course content. Practical tasks are performed and written-up well.
- 70%-100%:** All the previous to an excellent level. Demonstrates an ability to put a numerical solution into a context and assess whether such solutions are meaningful. Experimental work is carried out to a high standard. Reports are clear and insightful with the student demonstrating an ability to reflect on the strengths and weaknesses of the experimental techniques used.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		246

Supplementary Material

- Johnson, K. *_Physics for You_*. UK: Stanley-Thornes, 2011.
- O'Regan, D. *_Real-World Physics_*. Dublin: Folens, 2000.
- Squires, G.L. *_Practical Physics_*. UK: Cambridge University Press, 2008.

Requested Resources

- SCIENCE LAB: Physics